
Applications of Topology to the Detection of Ventricular Tachycardia

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Introduction

“In Einstein’s general relativity, the structure of space can change but not its topology. Topology is the property of something that doesn’t change when you bend it or stretch it as long as you don’t break anything.”

-Edward Witten

- Tachyarrhythmias are rapid heartbeats that are abnormal. They often cause sudden cardiac arrest and even death, so they must be diagnosed quickly.
- Most sudden cardiac deaths (about 50%) caused by VT or VF, about 300,000 deaths each year in U.S. (Compton)
- Most doctors try to detect symptoms manually, which is prone to error, with over 40% of doctors claiming they often lose focus due to fatigue. (Gregory)
- Methods in the current literature can only detect general categories of Arrhythmia, and not the specific type of Arrhythmia or its location. (Dindin et al.)
- Modern tests that give more detailed results cost hundreds to thousands of dollars and are often not conclusive.
- Some applications of machine learning have been done, but require excessive preprocessing and a large number of channels for filtering, and an unnecessary number of parameters. (Dindin et al.)

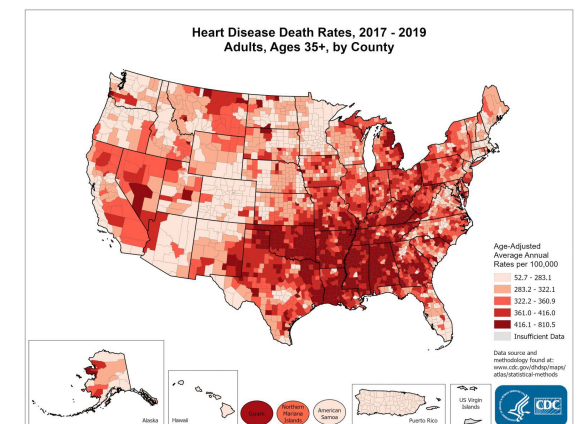


Fig.1. Map of Heart Disease Death Rates, Source: CDC

Framework

Caused by the creation of a reentry circuit, which changes heartbeat pace



Ventricular Tachycardia - heart continues to beat regularly, but the ventricles beat too quickly so the heart cannot fill with blood.

Caused by one or multiple reentry systems that make regular waves



Ventricular Flutter - extremely unstable, one of the most indiscernible heart conditions, ventricles beat chaotically, heart cannot fill with blood

Caused by multiple reentry systems that move in the ventricles and constantly change



Ventricular Fibrillation - most common deadly tachyarrhythmia. Ventricles quiver and twitch, not expanding and contracting, and cannot pump blood

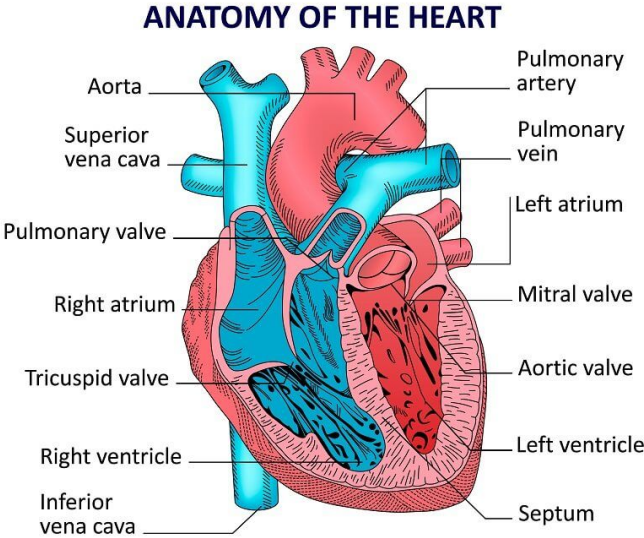


Fig.1. Anatomy of Heart, Source: Biology Dictionary

Framework

Topology: Topology studies the properties of a geometric space under deformations.

Metric Space:

A metric space consists of a set of points M and a distance function $d : M \times M \rightarrow \mathbb{R}_{\geq 0}$. This function d must satisfy three criteria.

- (i) d is symmetric ($\forall a, b \in M, d(a, b) = d(b, a)$)
- (ii) $\forall a, b \in M, d(a, b) \geq 0$, with equality if and only if $a = b$. (d is positive definite)
- (iii) d satisfies the triangle inequality ($\forall a, b, c \in M, d(a, b) \leq d(a, c) + d(c, b)$)

Time series are a type of metric space.

Metric Space: Time Series of ECG Data

Deformations: Higher dimensional embeddings, persistent homologies of embeddings

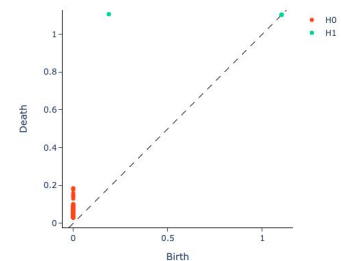
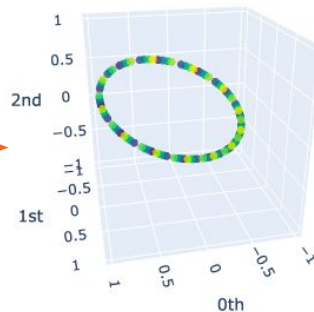
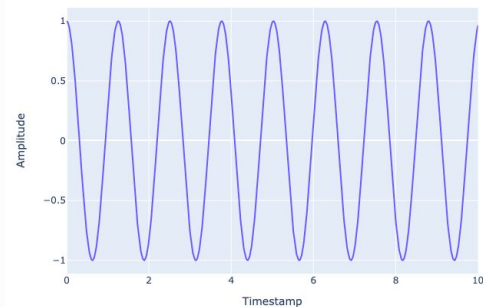


Fig.2. Example of time series, by author

Fig.3. Example of embedding, by author

Fig.4. Example of persistent homology, by author

Framework

Takens' Embedding Theorem:

Denote a time series by $\{x_t : t \in T\}$. Choose a time delay of ρ and an embedding dimension of size d . Then the time series embedding will become a set of d dimensional vectors, specifically, if we represent the time series as a function $f : A \rightarrow B$ and denote the embedding by T , then $T(f) = \{[f(t), f(t - \rho), f(t - 2\rho), \dots, f(t - (d - 1)\rho)]^T | t \in A\}$. Taken's Theorem states that the embedding of the original time series is *topologically equivalent* to the attractor generated by this embedding, known as a *delay embedding*.

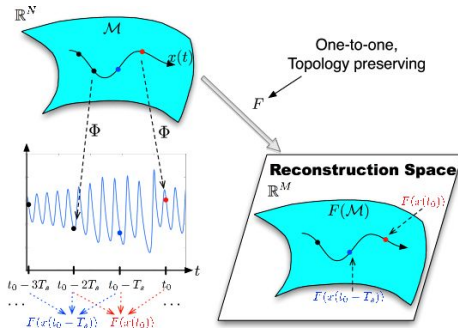


Fig.6. Embedding conserves topology, by Nuit Blanche

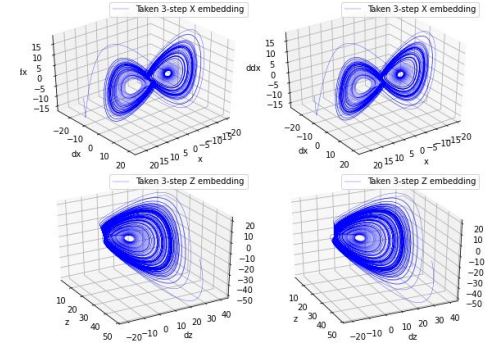


Fig.7. Lorenz Embedding, by Lutz Lehmann

Method of Mutual Information:

To determine the best ρ we first calculate the maximum $x_{\max}$ and $x_{\min}$ values of the time series, and divide the interval $[x_{\min}, x_{\max}]$ into a large number of bins. We let p_k be the probability that an element of the time series is in the k th bin and $p_{j,k}$ be the probability that x_i is in the j th bin while $x_{i+\rho}$ is in the k th bin. Then the mutual information is defined as $I(\rho) = - \sum_{j=1}^{n_{\text{bins}}} \sum_{k=1}^{n_{\text{bins}}} p_{j,k}(\rho) \log \frac{p_{j,k}(\rho)}{p_j p_k}$. The first minimum of $I(\rho)$ gives the optimal time delay since there we get the most information by adding $x_{i+\rho}$.

Framework

Filtration:

For an index set I , a filtration is a sequence of simplicial complexes, $\{\mathbb{K}_t\}_{t \in I}$, satisfying $\mathbb{K}_{t_1} \subseteq \mathbb{K}_{t_2}$ whenever $t_1 \leq t_2$.

Filtration of Simplicial Complexes:

$\forall i \leq j$, we can construct a mapping $f_p^{i,j} : H_p(\mathbb{K}_i) \rightarrow H_p(\mathbb{K}_j)$ for all dimensions p . The filtration can then be thought of as chain of homology groups of each of the respective simplicial complexes as follows:

$$0 = H_p(\mathbb{K}_0) \rightarrow H_p(\mathbb{K}_1) \rightarrow \cdots \rightarrow H_p(\mathbb{K}_n).$$

As we move through the chain, some components are lost while at other points, some components are gained.

The images of the original mapping $f_p^{i,j}$ are known as the *persistent homology groups*, and the components of these groups form the **persistent homology** of the filtration.

Notions of Birth and Death:

Let $\{\mathbb{K}_i\}_{i=0}^n$ be a filtration of simplicial complexes and let γ be a p th component in $H_p(\mathbb{K}_i)$. We say that γ is *born* at $\mathbb{K}_i$ if $\gamma \notin H_p^{i-1,i}$. Furthermore, if γ is born at $\mathbb{K}_i$ then it *dies* entering $\mathbb{K}_j$ if it merges with another hole as we go from $\mathbb{K}_{j-1}$ to $\mathbb{K}_j$, that is $f_p^{i,j-1}(\gamma) \notin H_p^{i-1,j-1}$ but $f_p^{i,j}(\gamma) \in H_p^{i-1,j}$.

Framework

Vietoris Rips-Complex and the Rips Filtration

Let $X = \{x_1, x_2, \dots, x_N\} \in \mathbb{R}^p$ be a point cloud and take some $\epsilon > 0$. The *Vietoris Rips complex* is a collection of all simplexes σ with vertices in X with $\text{diam}(\sigma) \leq 2\epsilon$, where $\text{diam}(\sigma)$ denotes the diameter of σ . $VR(X, \epsilon) := \cup_{q=0}^p \{q - \text{simplex}(\sigma) | \text{diam}(\sigma) \leq 2\epsilon, V(\sigma) \subseteq X\}$

Rips Filtration for an Embedding of Dimension 2:

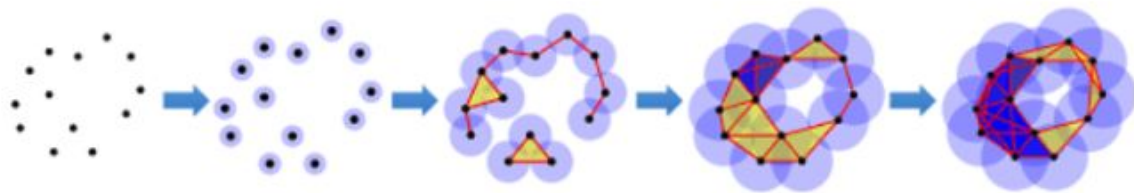


Fig.8. Example Embedding, Chi Seng Pun

Persistence Diagrams (of Persistent Homology Groups):

-A diagram which illustrates the birth and death times of components of pth homology groups.

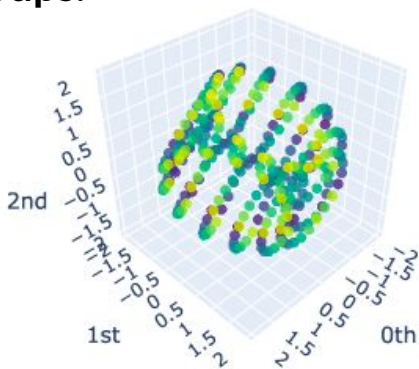


Fig.5. Example Embedding, by author

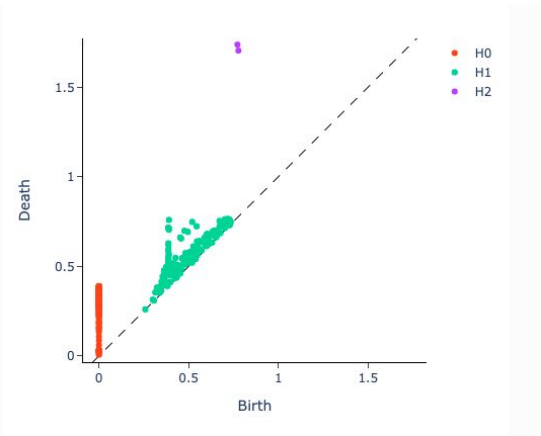


Fig.6. Example Persistence Diagram, by author

Findings

Healthy Control:

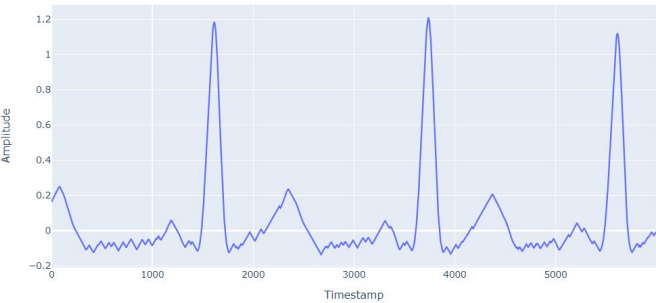


Fig.7. Control Original Data, by author

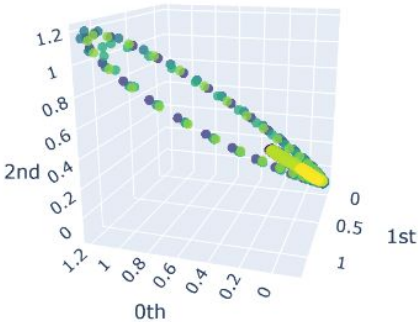


Fig.8. Control Embedding, by author

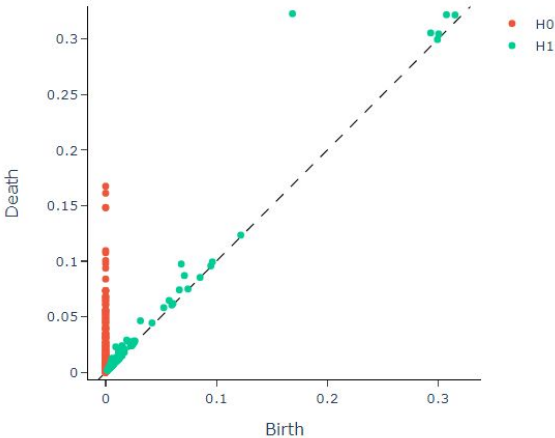


Fig.9. Control Persistence Diagram, by author

Ventricular Tachycardia:

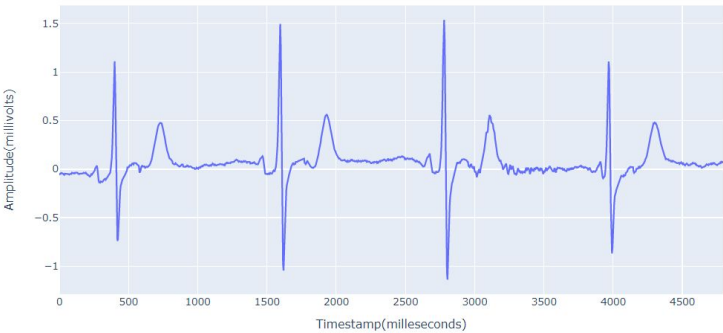


Fig.10. Tachycardia Original Data, by author

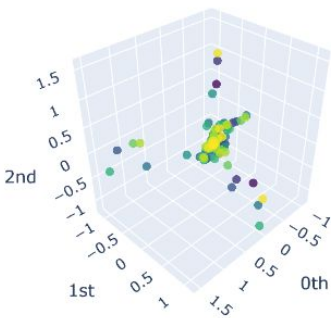


Fig.11. Tachycardia Embedding, by author

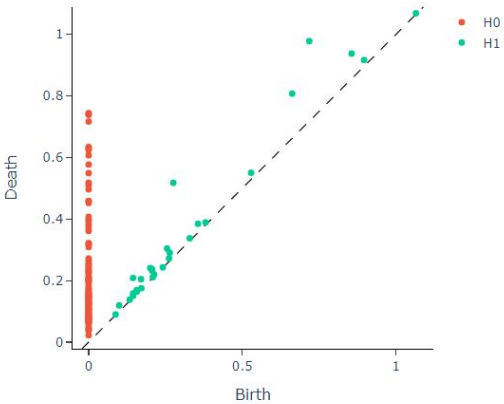


Fig.12. Tachycardia Persistence Diagram, by author

Findings

Ventricular Flutter

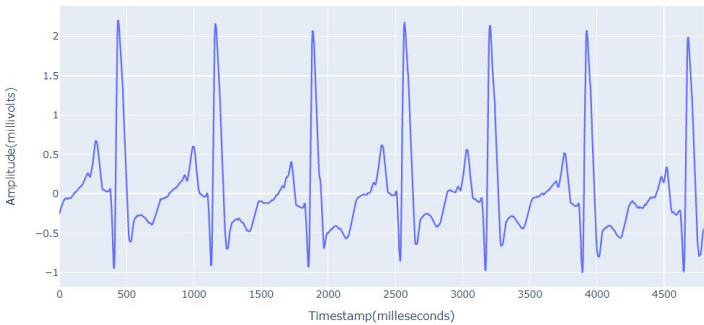


Fig.13. Original Fibrillation Data, by author

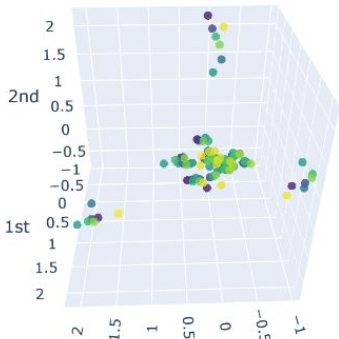


Fig.14. Fibrillation Embedding, by author

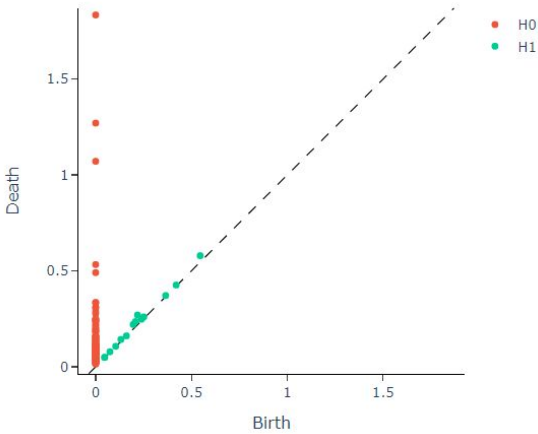


Fig.15. Fibrillation Persistence Diagram, by author

Ventricular Fibrillation

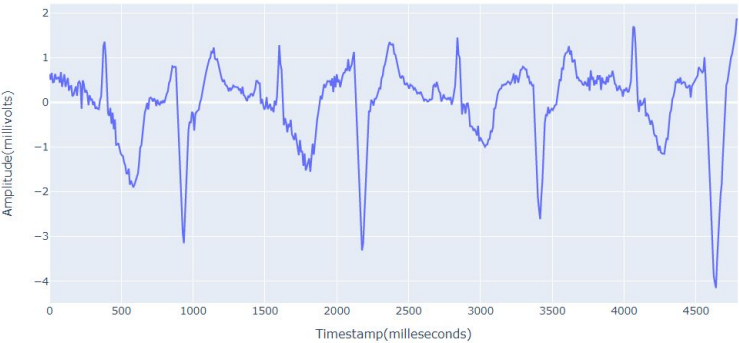


Fig.16. Original Flutter Data, by author

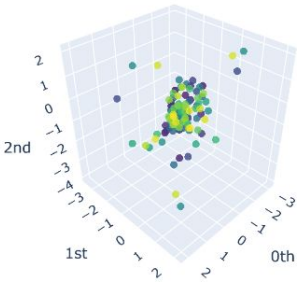


Fig.17. Flutter Embedding, by author

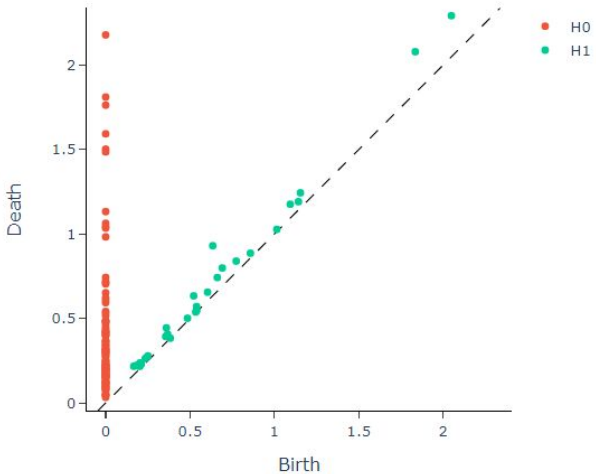


Fig.18. Flutter Persistence Diagram, by author

Findings

Results:

The three main novel steps we applied to the data in our research were:

- Takens' Embedding using Mutual Information on ECG data
- Rips Filtration using the Vietoris Rips Complex
- Generating Persistence Diagram using Scikit Learn Algorithms (Giotto TDA)

A key distinction can be seen between the healthy controls, tachycardia patients, flutter patients, and fibrillation patients by comparing the latest death time of their components. The trend is that healthy controls have a latest death time in the range 0.15-0.20, tachycardia patients have one in the range 0.8-1.0, flutter patients have one in the range 1.5-2.0, and fibrillation patients have one in the range 2.0-2.4.

Information on Datasets:

- All data was acquired from CU Ventricular Tachyarrhythmia database from PhysioNet.org, a collection of databases of free medical research data created and managed by the MIT Laboratory for Computational Physiology.
 - Contained ECG recordings and annotations of patients suffering from Ventricular Tachycardia, Ventricular Fibrillation, and Ventricular Flutter.
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Findings

Machine Learning Code for Takens' Embedding:

```
1 import pandas as pd
2 import numpy as np
3 import plotly.graph_objects as go
4 from gtdata.plotting import plot_point_cloud
5
6 from sklearn import datasets
7 from sklearn.cluster import DBSCAN
8 from sklearn.decomposition import PCA
9
10 df = pd.read_excel('C:\\Users\\jonat\\OneDrive\\Desktop\\Research\\Heart Rate\\Patient1.xlsx')
11 y_list1 = df['ECG'].tolist()
12
13 from gtdata.time_series import SingleTakensEmbedding
14
15 embedding_dimension = 3
16 maximum_time_delay = 3
17 stride = 5
18
19 embedder_periodic = SingleTakensEmbedding(
20     parameters_type="search",
21     time_delay=maximum_time_delay,
22     dimension=embedding_dimension,
23     stride=stride,
24 )
25
26 def fit_embedder(embedder: SingleTakensEmbedding, y: np.ndarray) -> np.ndarray: #Mutual Information
27     #Fits a Takens embedder by finding optimal time delay
28     y_embedded = embedder.fit_transform(y)
29     return y_embedded
30
31 y_periodic_embedded = fit_embedder(embedder_periodic, y_list1)
32
33 pca = PCA(n_components=3)
34 y_periodic_embedded_pca = pca.fit_transform(y_periodic_embedded)
```



Fig.23. Giotto-tda logo, by Giotto-tda

Fig.24. Taken's Embedding Code, by author

Machine Learning Code for Persistence Diagram:

```
1 from gtdata.homology import VietorisRipsPersistence
2
3 # 0 - for connected components, 1 - for loops, 2 - for voids
4 homology_dimensions = [0, 1, 2]
5
6 periodic_persistence = VietorisRipsPersistence(
7     homology_dimensions=homology_dimensions, n_jobs=
8 )
9 print("Persistence diagram for periodic signal")
10 periodic_persistence.fit_transform_plot(y_periodic_embedded)
```

Fig.25. Persistence Diagram Code, by author

Conclusions

Applications/Progress:

Our work offers an alternative approach to methods in the current literature to detect Ventricular Tachycardia and related heart conditions. After cross validation, our work is seen to be competitive/perform even better than current algorithms in the literature. This provides a cost-effective method that can be applied to live ECG data, and demonstrates how the 0 dimensional components of persistent homology embeddings of ECG data can drastically help save the lives of the millions of people who suffer from heart disease.

Future Work/Limitations:

Future work could focus on studying higher dimensional topologies. This is one of the limitations of our work, because topologies above 0 dimensional topologies are not accounted for.

References:

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